Repository Structure

Complete overview of the RAG tutorial repository.

■ Directory Layout

```
rag_tutorial/
■
■■■ ■ README.md                      # Main introduction and navigation
■■■ ■ QUICKSTART.md                  # 5-minute setup guide
■■■ ■ requirements.txt               # Python dependencies
■■■ ■ .env.example                   # Environment variables template
■■■ ■ demo.py                        # Quick demonstration script
■
■■■ ■ lessons/                       # Step-by-step tutorials
■   ■■■ 01-introduction-to-rag.md
■   ■■■ 02-understanding-embeddings.md
■   ■■■ 03-vector-databases-retrieval.md
■   ■■■ 04-language-models-generation.md
■   ■■■ 05-building-simple-rag.md
■   ■■■ 06-advanced-rag-techniques.md
■
■■■ ■ examples/                      # Working code examples
■   ■■■ 01-simple-qa/
■   ■   ■■■ simple_qa.py
■   ■   ■■■ README.md
■   ■■■ 02-document-chat/
■       ■■■ document_chat.py
■       ■■■ README.md
■
■■■ ■■ utils/                         # Reusable utility functions
■   ■■■ __init__.py
■   ■■■ embeddings.py                # Embedding operations
■   ■■■ retrieval.py                 # Vector search
■   ■■■ generation.py                # LLM generation
■
■■■ ■ data/                          # Sample data for testing
    ■■■ sample_rag_article.txt
```

■ Learning Path

For Complete Beginners

1. **Start Here**: Read README.md for overview
2. **Quick Setup**: Follow QUICKSTART.md
3. **Run Demo**: Execute `python demo.py` to see RAG in action
4. **Learn Concepts**: Read lessons 1-4 in order
5. **Build System**: Follow lesson 5 to build your first RAG system
6. **Try Example**: Run `examples/01-simple-qa/simple_qa.py`
7. **Go Advanced**: Study lesson 6 for production techniques

For Experienced Developers

1. **Skim**: README.md and QUICKSTART.md
2. **Setup**: `pip install -r requirements.txt`
3. **Review**: Lessons 3-4 for implementation details
4. **Code**: Study `examples/` and `utils/` code
5. **Advanced**: Jump to lesson 6 for techniques
6. **Build**: Create your own system using utilities

■ Lesson Summaries

Lesson 1: Introduction to RAG

- What RAG is and why it matters
- Problems it solves (hallucinations, knowledge cutoff)
- Three-step process: Index, Retrieve, Generate
- Real-world applications

Lesson 2: Understanding Embeddings

- What embeddings are (numerical text representations)
- How they capture semantic meaning
- Popular embedding models
- Cosine similarity for measuring relevance

Lesson 3: Vector Databases & Retrieval

- Why specialized databases are needed

- Approximate nearest neighbor (ANN) algorithms
- Popular vector databases (Chroma, Pinecone, etc.)
- Document chunking strategies
- Retrieval techniques

Lesson 4: Language Models & Generation

- How LLMs work at a high level
- Effective prompting for RAG
- Choosing and using LLMs
- Handling common issues

Lesson 5: Building a Simple RAG System

- Complete implementation walkthrough
- Document loading and chunking
- Building the RAG pipeline

- Command-line interface
- Testing and evaluation

Lesson 6: Advanced RAG Techniques

- Semantic chunking
- Hybrid search (semantic + keyword)
- Re-ranking for better results
- Query expansion and multi-query
- Production optimizations

■ Example Summaries

Example 1: Simple QA

File: `examples/01-simple-qa/simple_qa.py`

Minimal RAG system that: - Loads facts into vector database - Answers questions using retrieval - Works without OpenAI API key - Perfect for understanding basics

Run: `python examples/01-simple-qa/simple_qa.py`

Example 2: Document Chat

File: `examples/02-document-chat/document_chat.py`

Interactive chat system that: - Loads text documents - Chunks them intelligently - Maintains conversation history - Cites sources in answers - Requires OpenAI API key

Run: `python examples/02-document-chat/document_chat.py`

■■ Utility Modules

`utils/embeddings.py`

- `EmbeddingManager`: Manage embedding models
- `embed_text()`: Convert text to vectors
- `compute_similarity()`: Calculate text similarity

Example:

```
from utils import EmbeddingManager
embedder = EmbeddingManager()
vector = embedder.embed("Hello world")
```

`utils/retrieval.py`

- `Retriever`: Basic vector search
- `HybridRetriever`: Advanced retrieval with filtering
- Search with thresholds and MMR

Example:

```
from utils import Retriever
retriever = Retriever()
retriever.add_documents(["doc1", "doc2"])
results = retriever.search("query")
```

`utils/generation.py`

- `Generator`: LLM text generation
- `PromptTemplate`: Reusable prompt templates
- RAG-specific generation helpers

Example:

```
from utils import Generator
generator = Generator(api_key="...")
answer = generator.generate_with_context(query, context)
```

■ Sample Data

`data/sample_rag_article.txt`

Comprehensive article about RAG covering: - Introduction and motivation - How RAG works - Key components - Best practices - Applications - Challenges

Use this for testing your RAG systems!

■ Quick Start Commands

```
# Setup
source .venv/bin/activate
pip install -r requirements.txt

# Run demo
python demo.py

# Run example 1 (no API key)
python examples/01-simple-qa/simple_qa.py

# Run example 2 (needs API key)
export OPENAI_API_KEY="your-key"
python examples/02-document-chat/document_chat.py

# Test utilities
python utils/embeddings.py
python utils/retrieval.py
```

■ File Purposes

File	Purpose
`README.md`	Main entry point, overview
`QUICKSTART.md`	Fast setup guide
`requirements.txt`	Python dependencies

`.env.example`	Environment variable template
`demo.py`	Quick demonstration
`lessons/*.md`	Educational content
`examples/*/`	Working code samples
`utils/*.py`	Reusable utilities
`data/*.txt`	Sample documents

■ Where to Go Next

- **New to RAG?** → Start with Lesson 1
- **Want to code?** → Run `demo.py` and examples
- **Need utilities?** → Check `utils/` modules
- **Building app?** → Study Lesson 5
- **Production ready?** → Read Lesson 6

Happy Learning! ■